PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

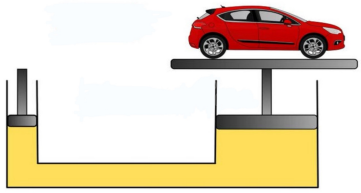
MIDTERM EXAM
3/24/26

Name _____
NetID _____

BEFORE YOU START…

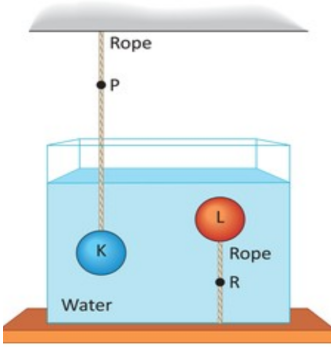
On the exam, blue book, and answer sheet, write:
- your FULL name (as listed by the university registrar)
- your NetID (i.e. your NYU email minus "@nyu.edu")

1. In the hydraulic lift shown, a car that weighs 13,500 N is held at rest on the piston at the right by a force applied to the smaller piston at the left. Both pistons are currently at the same altitude above the bottom of the apparatus, and the diameter of the piston on the right is 3 times the diameter of the one on the left. How large is the force being applied to the left piston?



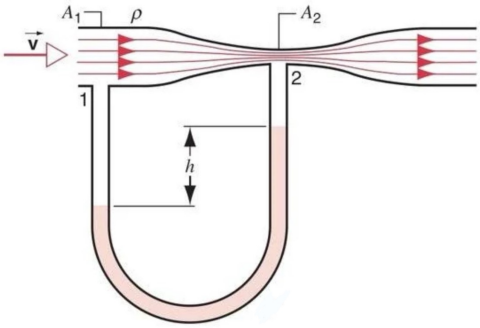
a) 1500 N b) 1600 N c) 1700 N d) 1800 N e) 1900 N

2. Two identically shaped spheres K and L are fully submerged in a tank of water and connected to light inextensible ropes as shown. The tension at point R is twice the weight of L, and the tension at point P is 1/9 the weight of K. What is the ratio m_K / m_L of the masses of the spheres?



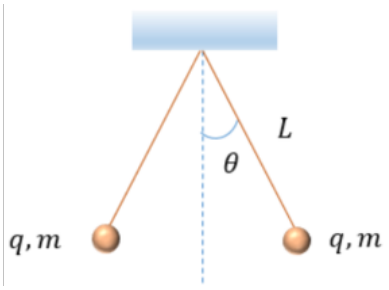
a) 45/14 b) 36/11 c) 27/8 d) 18/5 e) 9/2

3. A fluid of density ρ enters a tube of cross-sectional area A_1 at speed v = 1.1 m/s. It passes through a narrower section of cross-sectional area $A_2 = A_1 / 4$ before entering another section of cross-sectional area A_1 (ignore any vertical pressure variation within the tube). A **water-filled** manometer (U-tube) connects the wide and narrow sections of the tube and shows a difference h = 5 cm in the water level between its two arms. What is the approximate value of ρ? Use g ≈ 10 m/s^2 to simplify the arithmetic.



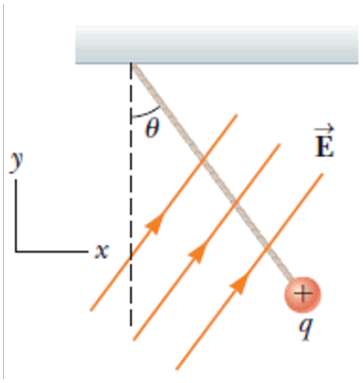
a) 18.5 kg/m^3 b) 23.1 kg/m^3 c) 29.6 kg/m^3
d) 39.4 kg/m^3 e) 55.1 kg/m^3

4. Two small identical charged spheres hang from cords of equal length L = 35 cm, each of which makes an angle of 15 degrees with the vertical as shown. If the mass of each sphere is m = 50 grams, what is the magnitude of the charge on each sphere?



a) 1.29 μC b) 0.692 μC c) 3.30 μC d) 1.88 μC e) 2.47 μC

5. A positively charged cork ball of mass m is suspended on a light string in the presence of a uniform electric field, as shown in the figure. When **E** = (A**î** + B**ĵ**) N/C, where A and B are positive numbers, the ball is in equilibrium at the angle θ. Gravity points down as usual. Which of the following is a correct expression for the charge *q* on the ball?



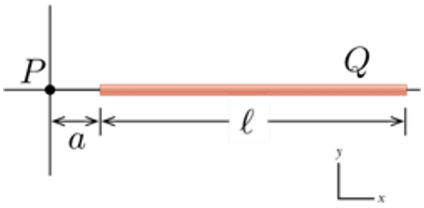
a) $\frac{Ag}{cot(\theta)+B}$ b) $\frac{mg}{Bcot(\theta)+A}$ c) $\frac{mg}{Acot(\theta)+B}$

d) $\frac{mg}{Btan(\theta)+A}$ e) $\frac{mg}{cot(\theta)+1}$

6. An electron is shot perpendicularly toward the center of a large negatively charged plate from a distance of 10 m. If the initial velocity of the electron is 3×10^7 m/s and the field produced by the plate is E = 400 N/C, how close to the plate does the electron get before stopping?

a) 12.7 m b) 4.88 m c) 3.59 m d) 5.73 m e) 1.46 m

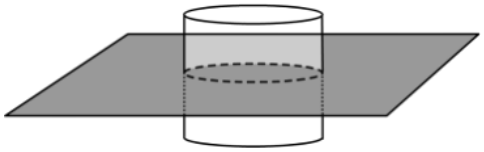
7. Point *P* is located a distance a from the left end of a uniformly charged thin rod. The rod has total charge *Q* and length $\ell = 5.4a$. Which of the following gives the correct electric field at point *P*?



a) $-\frac{5\,k\,Q}{26\,a^2}\hat{\imath}$ b) $-\frac{5\,k\,Q}{32\,a^2}\hat{\imath}$ c) $-\frac{5\,k\,Q}{38\,a^2}\hat{\imath}$

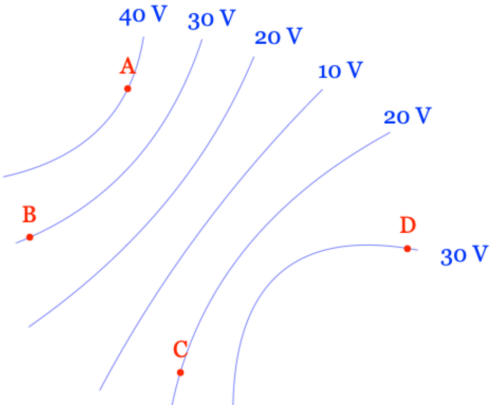
d) $-\frac{5\,k\,Q}{44\,a^2}\hat{\imath}$ e) $-\frac{k\,Q}{a^2}\hat{\imath}$

8. A closed cylindrical surface is oriented such that its bases are parallel to an infinite plane of uniform surface charge density as shown in the figure. The area of each base is 0.25 m^2 and the height of the cylinder is 0.8 m. If the electric flux through the entire closed cylindrical surface is 280 $N \cdot m^2/C$, what is the surface charge density of the plane?



a) 9.91 nC/m^2 b) 3.89 nC/m^2 c)6.73nC/m^2
d) 5.31 nC/m^2 e) 8.50 nC/m^2

9. The diagram shows some equipotential curves in a region of space when the potential ranges from 10V to 40V. What is the work that must be done to take a positive 4.3 nC point charge that is initially at rest at point D and place it at rest at point A? (1 nC = 10^{-9} C)



a) 21 nJ b) 32 nJ c) 43 nJ
d) 54 nJ e) 0

10. A neutral spherical conductor has a non-concentric smaller spherical cavity that has a positive point charge +q at its center (*the image is NOT drawn to scale!*). The radius of the cavity is *b*. The distance between the center of the larger conductor and the center of the cavity is 3*b*. The radius of the larger spherical conductor is 17*b*. Take the electric potential to be zero infinitely far from this arrangement. What is the electric potential on the curved walls of the cavity?



a) $kq\ /\ b$ b) $kq\ /\ (11\,b)$ c) $kq\ /\ (13\,b)$
d) $kq\ /\ (15\,b)$ e) $kq\ /\ (17\,b)$

PH2023 Equation Sheet

$V_{\text{sphere}} = \frac{4}{3}\pi R^3$	$A_{\text{sphere}} = 4\pi R^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2/\text{N}\cdot\text{m}^2$	$k_e = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
$\mu_0 = 4\pi \times 10^{-7} \text{T}\cdot\text{m/A}$	$e = 1.60 \times 10^{-19}$ C	$m_e = 9.11 \times 10^{-31}$ kg	$m_p = 1.67 \times 10^{-27}$ kg
$P_{atm} = 1.013 \times 10^5$ Pa	1 eV $= 1.60 \times 10^{-19}$ J	$\rho_{\text{water}} = 1000 \text{ kg/m}^3$	

$$
\begin{aligned}
x &= x_0 + v_0 t + \frac{1}{2}at^2 \\
F_c &= \frac{mv^2}{r} \\
\boldsymbol{\tau} &= \mathbf{r} \times \mathbf{F} \\
P &= F/A \\
P &= P_0 + \rho g h \\
F_B &= \rho_{\text{fluid}} g V \\
Av &= \text{const} \\
P + \frac{1}{2}\rho v^2 + \rho g y &= \text{const} \\
\mathbf{F}_{12} &= \frac{k q_1 q_2}{r_{12}^2}\hat{r}_{12} \\
\mathbf{E} &= \frac{kQ}{r^2}\hat{r} \\
E &= \frac{\sigma}{2\epsilon_0}, \quad E = \frac{2k\lambda}{r} \\
\mathbf{F} &= q\mathbf{E} \\
\boldsymbol{\tau} &= \mathbf{p} \times \mathbf{E} \\
U &= -\mathbf{p} \cdot \mathbf{E} \\
V &= \frac{kQ}{r} \\
E_x &= -\frac{\partial V}{\partial x} \\
\Delta V &= -\int_A^B \mathbf{E} \cdot d\boldsymbol{\ell} \\
\Delta V &= -Ed \\
U &= k\frac{q_1 q_2}{r} \\
U &= qV \\
R &= \rho\frac{l}{A} \\
\rho &= \rho_{20}[1 + \alpha(T - 20^\circ)] \\
I &= nqv_d A \\
\mathcal{P} &= I^2 R \\
\oint \mathbf{E} \cdot d\mathbf{A} &= \frac{Q_{\text{in}}}{\epsilon_0} \\
Q &= CV \\
C_0 &= \epsilon_0\frac{A}{d} \\
C &= \kappa C_0 \\
U &= \frac{1}{2}QV = \frac{Q^2}{2C} = \frac{1}{2}CV^2 \\
C_{\text{eq}} &= C_1 + C_2 + C_3 + \cdots \\
\frac{1}{C_{\text{eq}}} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
V &= IR \\
R_{\text{eq}} &= R_1 + R_2 + R_3 + \cdots \\
\frac{1}{R_{\text{eq}}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \\
\mathbf{F} &= q\mathbf{v} \times \mathbf{B} \\
\mathbf{F} &= I\boldsymbol{\ell} \times \mathbf{B} \\
r &= \frac{mv}{qB} \\
T &= \frac{2\pi m}{qB} \\
\mathcal{E} &= -N\frac{d\phi_B}{dt} \\
\mathcal{E} &= Blv \\
L &= \frac{N\phi_B}{I} \\
L &= \frac{\mu_0 N^2 A}{l} \\
\mathcal{E}_L &= -L\frac{dI}{dt} \\
U &= \frac{1}{2}LI^2 \\
\phi_B &= \int \mathbf{B} \cdot d\mathbf{A} \\
\boldsymbol{\mu} &= NI\mathbf{A} \\
\boldsymbol{\tau} &= \boldsymbol{\mu} \times \mathbf{B} \\
U &= -\boldsymbol{\mu} \cdot \mathbf{B} \\
\oint \mathbf{B} \cdot d\boldsymbol{\ell} &= \mu_0 I_{in} \\
\mathbf{B} &= \frac{\mu_0 I}{4\pi}\int \frac{d\boldsymbol{\ell} \times \hat{r}}{r^2} \\
B &= \mu_0 n I \\
B &= \frac{\mu_0 I}{2\pi R} \\
B &= \frac{\mu_0 I}{2R} \\
E &= cB \\
u_E &= \frac{1}{2}\epsilon_0 E^2 \\
u_B &= \frac{B^2}{2\mu_0} \\
\omega &= \frac{1}{\sqrt{LC}} \\
Q(t) &= C\mathcal{E}(1 - e^{-t/RC}) \\
Q(t) &= Q_0 e^{-t/RC} \\
I(t) &= I_0 e^{-t/RC} \\
I(t) &= I_0 e^{-Rt/L} \\
I(t) &= \frac{\mathcal{E}}{R}(1 - e^{-Rt/L})
\end{aligned}
$$